

Influence of temperature, state of charge and state of health on the thermal parameters of lithium-ion cells: Exploring thermal behavior and enabling fast-charging

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ABSTRACT

The precise input of thermal parameters is essential for thermal simulation. Although constant thermal parameters are commonly used for parametrizing thermal modeling frameworks, extensive measurements indicate a significant dependence of thermal parameters on temperature, SOC and SOH. Therefore, this work summarizes experimental data and integrates determined operating point dependencies into a validated thermal-electrical-electrochemical modeling framework. Exploring the effect of variable thermal parameters, detailed effects on fast-charging and corresponding charging times are assessed.

It is found that the strong reduction in through-plane thermal conductivity due to aging can notably increase thermal inhomogeneity. Thus, heat dissipation is reduced and the thermal management has to be revised to prevent an increase in charging time of up to 3%. However, the operating point-dependent through-plane thermal conductivity has no significant effect on fast-charging for the analyzed pristine cylindrical lithium-ion cell. Furthermore, a temperature-dependent specific heat capacity definition considerably affects the thermal behavior of lithium-ion cells at extreme temperatures. While enabling a faster heating at low temperatures, a temperature-related current derating at high temperatures is delayed. Thus, a variable thermal parameter definition can lead to an increase in fast-charging capability of up to 3% due to the more precise modeling of the physical behavior of the cell.

1. Introduction

Lithium-ion batteries are considered a key driver for the energetic transition by electrifying mobility [1] and contributing towards smart grids [2]. Due to their high energy density, fast-charging capability and long service life, lithium-ion batteries are superior to traditional battery technologies [3]. In order to increase customer acceptance and ensure a widespread adoption, continuous development for increasing capacity, lifetime and safety is performed. Therefore, a thorough understanding of lithium-ion cells including simulation, design and precise state-estimation is required [4–6].

As lithium-ion cells are characterized by a complex interplay between thermal, electrical and electrochemical phenomena, various simulation frameworks exist [7,8]. While the frameworks can differ regarding their dimensionality, sub-model selection or software [9,10], a precise parametrization is of high importance for all modeling frameworks [11,12]. Besides many approaches for determining electrical model parameters [13], the input of thermal parameters is especially significant [14]. Being realized by measurements [15], analytical calculation [16] and utilizing literature values [17], all employed thermal parameters are generally assumed constant. However, extensive experimental data shows that thermal parameters can strongly vary over the operating range of lithium-ion cells [18]. Thus, the assumption of

Abbreviations: SOC, State of charge; SOH, State of health; TMS, Thermal management system; P2D, Pseudo-two-dimensional; EV-ARC, Extended volume accelerating rate calorimeter.

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Nomenclature

C_p	Specific heat capacity: Specific heat capacity of the full cell.
$C_{p,var}$	Variable specific heat capacity: Specific heat capacity depending on operating point.
I_{Cell}	Cell charging current: Maximum allowed charging current of the full cell.
I_{Max}	Maximum charging current: Maximum allowed cell element charging current.
$\lambda_{ }$	In-plane thermal conductivity: Thermal conductivity along cell layers.
$\lambda_{\perp}$	Through-plane thermal conductivity: Thermal conductivity across cell layers.
$\lambda_{\perp,var}$	Variable through-plane thermal conductivity: Thermal conductivity depending on operating point.
$\lambda_{\perp,aged}$	Aged through-plane thermal conductivity: Through-plane thermal conductivity in aged cells.
T_{Max}	Maximum temperature: Maximum allowed cell element temperature.
T_{Start}	Starting temperature: Starting temperature of a charging procedure.
$T_{Cool,On}$	Cooling activation temperature: Cell element temperature threshold which activates cooling.
T_{Hot}	Hotspot temperature: Temperature of the hottest cell element.
T_{Cold}	Coldspot temperature: Temperature of the coldest cell element.
U_{Cell}	Cell voltage: Voltage of the full cell.
$U_{Cell,Max}$	Maximum cell voltage: Maximum allowed voltage of the full cell.
U_{Ano}	Anode potential: Anode voltage vs Li/Li ⁺ .
$U_{Ano,Min}$	Minimum anode potential: Minimum allowed cell element anode potential.
X_{Ref}	Reference quantity: Constant reference for analyzing influence of variable thermal parameters.

constant thermal parameters is a simplification whereby corresponding uncertainties have not been explored and cannot be quantified.

Tranter et al. [19,20] developed a thermal simulation framework employing a pseudo-2-dimensional (P2D) electrochemical model for analyzing inhomogeneities among different cell designs. Assessing the influence of different tab configurations for large format cylindrical cells, thermal parameters were selected from literature. Moreover, Li et al. [4] present an equivalent circuit network framework for analyzing internal temperature distribution and cooling approaches for a 21700 cell. While individual material thermal parameters are selected from literature, jelly roll thermal parameters are obtained by analytical calculation. Another P2D electrochemical model was introduced by Sturm et al. [21,22]. Enabling the investigation of cell design, inhomogeneities and fast-charging capability, thermal parameters were again determined by selecting material parameters from literature and performing analytical approximation of jelly roll thermal parameters. Nevertheless, all presented simulations consider thermal parameters constant over the entire operating range. Even though the influence of temperature, state of charge (SOC) and state of health (SOH) on the thermal conductivity and specific heat capacity is clearly identified [18], the variation of thermal parameters at different operating points is neglected. Only Lin et al. [23] assessed the influence of a variable specific heat capacity on the heat generation inside lithium-ion cells. Measuring the specific heat capacity inside an extended-volume accelerating-rate-calorimeter (EV-ARC), significant deviations in heat generation between constant and variable specific heat capacity were

found. However, only a temperature-dependent specific heat capacity was considered whereby the influence of SOC and SOH as well as variable thermal conductivity were not analyzed. In addition, only influences on the heat generation were evaluated without considering fast-charging or aging in a sophisticated simulation of a full fast-charging cycle.

Thus, this article addresses the precise effect of operating point-dependent thermal parameters on the thermal simulation of lithium-ion cells first-ever. Therefore, a thoroughly validated and reliable thermal-electrical-electrochemical modeling framework is utilized. Collected experimental data is analyzed in order to determine the variation of thermal parameters at different operating points. Corresponding influences on the thermal simulation are assessed by analyzing charging time and related physical effects. In detail, this results in four subjects that will be analyzed in the following article:

1. Implementation of operating point-dependent thermal conductivity into the thermal simulation
2. Implementation of operating point-dependent specific heat capacity into the thermal simulation
3. Effect of operating point-dependent thermal parameters on thermal behavior and thermal management design
4. Effect of operating point-dependent thermal parameters on fast-charging time

The remainder of this article is structured as follows: [Section 2](#) briefly presents the utilized thermal simulation framework as well as the sample cell. Furthermore, employed operating point-dependent thermal parameters will be discussed in detail. Next, the third section analyzes the results regarding the effect of operating point-dependent thermal parameters on the thermal simulation. Therefore, corresponding influences of the operating point-dependent thermal parameters on fast-charging times and inhomogeneities are described. Finally, [Section 4](#) summarizes this work and presents potential for future work.

2. Modeling

The following sections present the thermal-electrical-electrochemical model by Pegel et al. [24] employed in this work as well as corresponding adaptations of the thermal parameters. Therefore, the modeling framework itself and the sample cell are briefly presented. Following, experimental data of thermal conductivity and specific heat capacity from previous work are utilized for parametrizing the thermal model. Finally, analyzed cooling scenarios are described.

2.1. Thermal-electrical-electrochemical model

This work employs the thermal-electrical-electrochemical modeling framework as presented by Pegel et al. [24]. For simplicity, the term thermal simulation refers to a thermal-electrical-electrochemical simulation in the following. The utilized thermal simulation is based on an electrochemical P2D model and is characterized by spatial resolution, high flexibility and scalability. While the governing equations for P2D models are presented in detail in Ref. [25], the simulation framework is implemented in the MATLAB/Simulink environment and can analyze arbitrary cell diameter, cell heights and housing materials. Various cooling scenarios can be integrated whereby the simulation was extended by a 3-stage coolant flow controller [26]. The framework is utilized in the following to calculate a minimum charging time from 10% to 80% SOC considering all relevant cell parameters as described by Pegel et al. [24] in addition to a chosen starting temperature and cooling scenario. Thereby, the maximum charging current is controlled by a P-controller such that:

- $T_{Max} \leq 65^\circ\text{C}$
- $I_{Max} \leq 4.25\text{ C}$

- $2.8 \text{ V} \leq U_{\text{Cell}} \leq 4.2 \text{ V}$
- $0.02 \text{ V} \leq U_{\text{Ano}}$

whereby the C-rate is the ratio between the charging current and the nominal capacity. Being thoroughly validated and tested, the identical 3×3 segregation with 9 rotationally symmetric volume elements is analyzed due to the tradeoff between computational cost and precision. The tabless design sample cell with high-nickel NCM cathode, silicon-oxide/graphite anode and DMC:EC:EMC electrolyte identical to Ref. [24] is utilized. The cell has a nominal capacity of 10 Ah with voltage limits between 2.5 V and 4.2 V. Thus, electrical and thermal

model parameters remain unchanged whereby only the operating point dependence of the thermal parameters is added.

2.2. Operating point-dependent thermal parameters

The thermal parameters of the sample cell for the thermal simulation were extensively validated by Pegel et al. [24] through EV-ARC measurements and thermal step response analysis. Being $\lambda_{\parallel} = 25 \text{ Wm}^{-1}\text{K}^{-1}$, $\lambda_{\perp} = 1 \text{ Wm}^{-1}\text{K}^{-1}$ and $C_p = 875 \text{ Jkg}^{-1}\text{K}^{-1}$ for the jelly roll, thermal parameters are originally assumed constant. However, comprehensive measurement data on thermal parameters exists [27–30] and indicates a

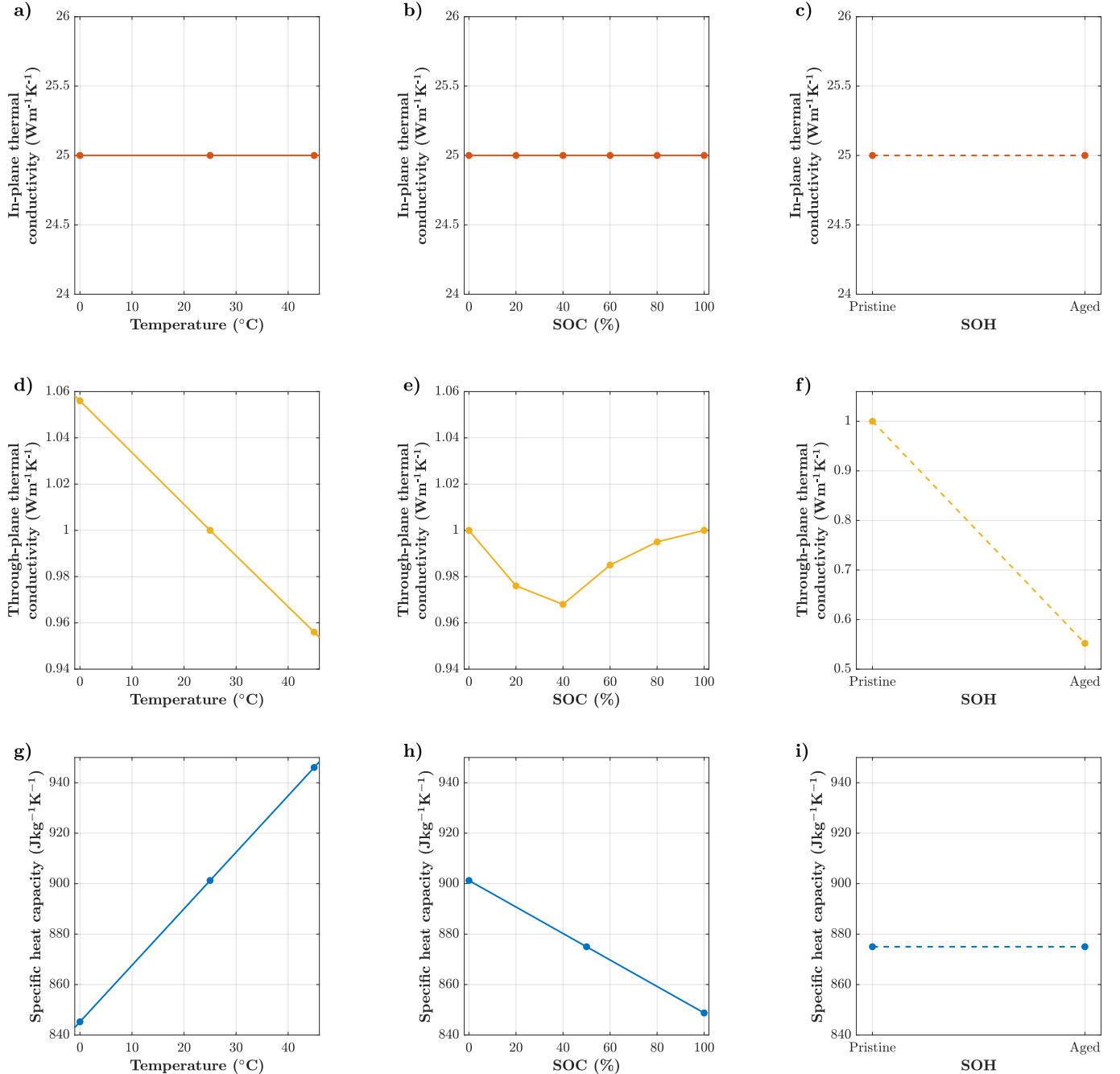


Fig. 1. Variable thermal parameters integrated into the thermal simulation for the sample cell [27–30]. (a) Temperature-dependent in-plane thermal conductivity for all SOCs. (b) SOC-dependent in-plane thermal conductivity for all temperatures. (c) SOH-dependent in-plane thermal conductivity for all temperatures and SOCs. (d) Temperature-dependent through-plane thermal conductivity at SOC = 0%. (e) SOC-dependent through-plane thermal conductivity at $T = 25^{\circ}\text{C}$. (f) SOH-dependent through-plane thermal conductivity for $T = 25^{\circ}\text{C}$ and all SOCs. (g) Temperature-dependent specific heat capacity at SOC = 0%. (h) SOC-dependent specific heat capacity at $T = 25^{\circ}\text{C}$. (i) SOH-dependent specific heat capacity for all temperatures and SOCs.

significant operating point dependence of thermal parameters. Therefore, original parameters are adopted and corresponding dependencies on temperature, SOC and SOH are integrated.

To begin with, the in-plane thermal conductivity of lithium-ion cells was thoroughly analyzed in previous work [29]. Utilizing a one-sided heating of sample cells with a resulting thermal gradient along the cell, the in-plane thermal conductivity of various cells can be experimentally determined. As the in-plane thermal behavior is dominated by the metallic current collectors, influences of temperature and SOC are negligible. As their thermal conductivity only varies by about one percent in the operating range of lithium-ion cells, the input of constant in-plane thermal conductivity with varying temperature and SOC is reasonable. In contrast, the influence of aging is considered relevant as the cell thickness increase reduces the in-plane thermal conductivity. However, the analyzed sample cell in this work is a cylindrical cell with aluminum housing. Aging-induced cell thickness increase for this specific cell is unexplored while other cylindrical cells do not experience similar cell thickness increase as pouch cells [31]. As a result, no effect of temperature and SOC on the in-plane thermal conductivity of lithium-ion cells can be found while the in-plane thermal conductivity of the analyzed cylindrical cell is not expected to be significantly affected by aging. Thus, a constant in-plane thermal conductivity of $\lambda_{||} = 25 \text{ Wm}^{-1}\text{K}^{-1}$ is implemented for all operating points as illustrated in Fig. 1.

The through-plane thermal conductivity of lithium-ion cells is subject to strong scattering with varying operating point as demonstrated in previous work [27,28]. By employing a vertical temperature gradient through one-sided heating across the cell layers, the through-plane thermal conductivity of various sample cells was analyzed and compared to analytical calculations under varying parametric conditions. Hence, the originally determined value of $\lambda_{\perp} = 1 \text{ Wm}^{-1}\text{K}^{-1}$ can be extended by a temperature dependence as shown in Fig. 1(d). A linear decrease in through-plane thermal conductivity of 10% between 0 °C and 45 °C as observed for the examined cells in [27] is implemented. Even though analyzed cells in previous work are pouch cells, the jelly roll of the sample cell in this work is expected to behave similar due to the negligible influence of the pouch foil. Similarly, the influence of SOC is added, too. Because all cells in [27] show a characteristic parabolic variation in through-plane thermal conductivity over increasing SOC, a similar through-plane thermal conductivity with 6 reference points is integrated for the sample cell. Depicted in Fig. 1(e), a moderate variation of 3% can be observed over the full SOC range. As previous work [27] demonstrated that no interaction effect of temperature and SOC on the through-plane thermal conductivity can be found, the parametric influences of temperature and SOC on the through-plane thermal conductivity can be analyzed individually. Fig. 1(f) depicts the influence of aging on the through-plane thermal conductivity. The through-plane thermal conductivity for aged cells is considered to be $\lambda_{\perp} = 0.5 \text{ Wm}^{-1}\text{K}^{-1}$. As extensively demonstrated in previous work [28], this is a worst case assumption and is used in this work to showcase extreme scenarios. While thermal transport is reduced, the temperature dependence of the through-plane thermal conductivity for aged cells decreases. As the SOC does not affect the through-plane thermal conductivity of strongly aged cells, it only depends on temperature.

Finally, the specific heat capacity of lithium-ion cells is subject to parametric influences, too. As extensively analyzed in previous work [30], temperature and SOC can have a significant influence on the specific heat capacity. Utilizing current pulses for controlled heating and cooling of sample cells, a flexible measurement approach for determining the specific heat capacity of a broad range of sample cells with varying parametric influences was employed. Since the specific heat capacity of the sample cell in this work was already thoroughly characterized by measurements in previous work, the influence of temperature and SOC can be directly integrated into the thermal simulation. Therefore, again the absolute value of $C_p = 875 \text{ Jkg}^{-1}\text{K}^{-1}$ determined by Pegel et al. [24] is not altered but a linear temperature-dependent increase of 0.25%/K is added as illustrated in Fig. 1(g). In contrast, the

specific heat capacity of the sample cell was measured to decrease with rising SOC. Hence, a linear decrease in specific heat capacity of 0.049%/SOC as shown in Fig. 1(h) is integrated into the thermal simulation. Again, previous work [30] indicated that no significant interaction effect between temperature and SOC on the specific heat capacity exists which allows for analyzing the factors individually. Finally, the specific heat capacity of lithium-ion cells in general was determined to not vary over lifetime [30,32]. Thus, Fig. 1(i) depicts the unchanged constant specific heat capacity of the analyzed sample cell in this work with varying SOHs.

In order to analyze the influence of individual as well as mixed operating point-dependent thermal parameters, six different parameter combinations are assessed in detail in the following. Summarized in Table 1, the selected combinations of in-plane thermal conductivity, through-plane thermal conductivity and specific heat capacity can be observed. As described above, the in-plane thermal conductivity is assumed constant over the full operating range and thus is not subject to any parametric influence in any of the analyzed parameter combinations. To begin with, the first parameter combination considers entirely constant thermal parameters and is considered the reference case in the following. Next, parameter set II describes the case of only variable through-plane thermal conductivity. In the following, variable characterization relates to the dependence of a thermal parameter on temperature and SOC. Parameter set III considers the aged through-plane thermal conductivity. In the following, aged through-plane thermal conductivity is defined to be reduced by 50% compared to the value for pristine cells. It is considered to only depend on temperature as described in the previous section. Furthermore, the case of only variable specific heat capacity is investigated with the fourth parameter set. As the variable thermal parameters are not expected to occur isolated, the physically intuitive combination of variable through-plane thermal conductivity and variable specific heat capacity is considered as depicted in parameter set V. Finally, the thermal parameter combination of an aged cell is examined with aged through-plane thermal conductivity and variable specific heat capacity in set VI. It has to be mentioned that only thermal parameters are adjusted to aging. Even though several cell parameters as inner resistance and capacity are also influenced by aging, for means of simplicity only the through-plane thermal conductivity is varied with sinking SOH. Furthermore, the linear effect of temperature displayed in Fig. 1 is extrapolated when the cell temperature reaches extreme temperature regions outside of the depicted temperature range. Even though the thermal parameters of the analyzed cell were not measured in described extreme temperature regions, measurements on material and cell level indicate a linear relationship throughout the thermal operating range of lithium-ion cells [33,34].

2.3. Cooling scenarios

Since the cooling approach has significant influence on the thermal behavior of lithium-ion cells, different cooling scenarios are analyzed in the following. Being no cooling, bottom cooling and side cooling, the examined concepts especially differ in cooling intensity as well as resulting temperature homogeneity.

First, no cooling represents the scenario of non-existing cooling whereby adiabatic boundary conditions at the cell surface are assumed.

Table 1
Analyzed thermal parameter combinations for the thermal simulation.

Set	Description	$\lambda_{ }$	$\lambda_{\perp}$	C_p
I	$\lambda_{\perp, \text{const}}$ & $C_{p, \text{const}}$	const.	const.	const.
II	$\lambda_{\perp, \text{var}}$ & $C_{p, \text{const}}$	const.	f(T, SOC)	const.
III	$\lambda_{\perp, \text{aged}}$ & $C_{p, \text{const}}$	const.	$f_{\text{red}}(T)$	const.
IV	$\lambda_{\perp, \text{const}}$ & $C_{p, \text{var}}$	const.	const.	f(T, SOC)
V	$\lambda_{\perp, \text{var}}$ & $C_{p, \text{var}}$	const.	f(T, SOC)	f(T, SOC)
VI	$\lambda_{\perp, \text{aged}}$ & $C_{p, \text{var}}$	const.	$f_{\text{red}}(T)$	f(T, SOC)

As demonstrated by Pegel et al. [24], no cooling is the suitable use case for charging with low starting temperatures. Because the critical temperature limit is not reached for cold starting temperatures, cooling would increase the charging time due to increased anodic overpotential. Hence, only starting temperatures below 10 °C do not demand for a cooling to achieve minimum fast-charging times for the considered sample cell. Therefore, the influence of variable thermal parameters on fast-charging with no cooling is only analyzed for $T_{\text{Start}} = 20$ °C and $T_{\text{Start}} = 10$ °C in the following.

Second, bottom cooling is an approach whereby a cooling plate is located on the bottom of the cell. Being frequently utilized in the automotive industry due to the compact size and straightforward spatial integration [35,36], it induces a significant axial temperature gradient. It is assumed that no heat exchange occurs along the circumferential dimension and the top. Furthermore, the heat transfer coefficient is considered to be $800 \text{ Wm}^{-2}\text{K}^{-1}$ as described by Pegel et al. [24] whereby the cell bottom is in full contact with the cooling plate.

Lastly, the test case of side cooling describes a cell circumferentially immersed in cooling fluid as illustrated in Fig. 2 (c). Leading to a very high cooling surface, a heat transfer coefficient of $400 \text{ Wm}^{-2}\text{K}^{-1}$ is assumed [37]. The cooling fluid is determined to be only in contact with the circumferential surface of the cell such that the top and bottom of the cell are considered adiabatic. Due to the large and evenly distributed cooling surface, a high cooling power with a relatively homogeneous temperature distribution is realized.

In order to analyze the influence of operating point-dependent thermal parameters on the thermal simulation, different starting temperatures are analyzed. While always examining the fast-charging of the sample cell from 10% to 80% SOC, starting temperatures of 20 °C and 10 °C are selected for no cooling as previously described. Furthermore, starting temperatures of 5 °C, 0 °C, 25 °C and 35 °C are assessed for bottom and side cooling. Higher temperatures are not considered due to the unlikely occurrence of such high starting temperatures. The cooling fluid temperature is assumed to be constantly equal to the initial starting temperature of the sample cell.

3. Results and discussion

The upcoming sections present the thermal simulation results in detail. Therefore, one exemplary charging procedure is described in detail. Following, the influence of each thermal parameter on fast-charging is presented. Correlating the physical behavior of lithium-ion cells to achieved simulations, the effect of variable thermal conductivity and variable specific heat capacity at different operating conditions is illustrated. Finally, an overview of achievable fast-charging times at various parameter combinations is given.

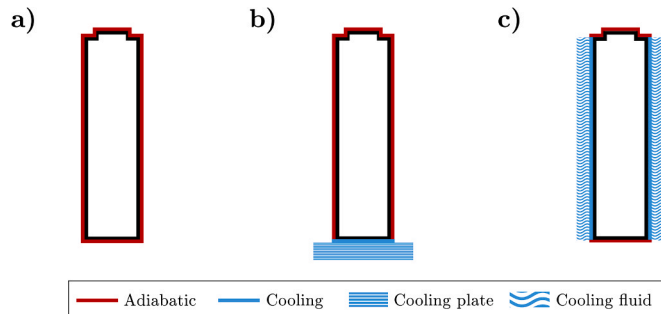


Fig. 2. Analyzed cooling scenarios. (a) No cooling with fully adiabatic boundary conditions. (b) Bottom cooling with bottom cooling plate whereby the lateral surface and top are considered adiabatic. (c) Side cooling with circumferential cooling fluid whereby the top and the bottom are considered adiabatic.

3.1. Exemplary charging procedure

To begin with, the charging procedure from 10% to 80% SOC with bottom cooling and a starting temperature of $T = 35$ °C is presented in detail. In order to understand the underlying control mechanisms as well as the thermal behavior of the lithium-ion cell, variable through-plane thermal conductivity and variable specific heat capacity are employed inside the simulation environment. By analyzing different charging currents, temperatures and anode voltages in detail, the governing physical cell limits can be understood in detail. Visualized in Fig. 3(a), the cell voltage and the anode voltage of each of the 9 volume elements can be observed. Furthermore, the temperature of the coldest and hottest volume element together with the average charging current is collected in Fig. 3(b). It is evident, that the maximum charging current of 4.25 C is the limiting factor for the first 330 s since neither the anode potential is below the limit of 20 mV and temperature is not critical, too. In order to prevent a thermal limitation of the charging current by exceeding the maximum temperature limit, cooling is initially activated and stays active for the entire charging procedure. At 330 s, the charging current is suddenly reduced to not exceed the maximum temperature limit of $T_{\text{Max}} = 65$ °C. Following, the charging current is controlled such that it is maximized while keeping the temperature of the hottest element at 65 °C. Hence, the maximum temperature is the governing

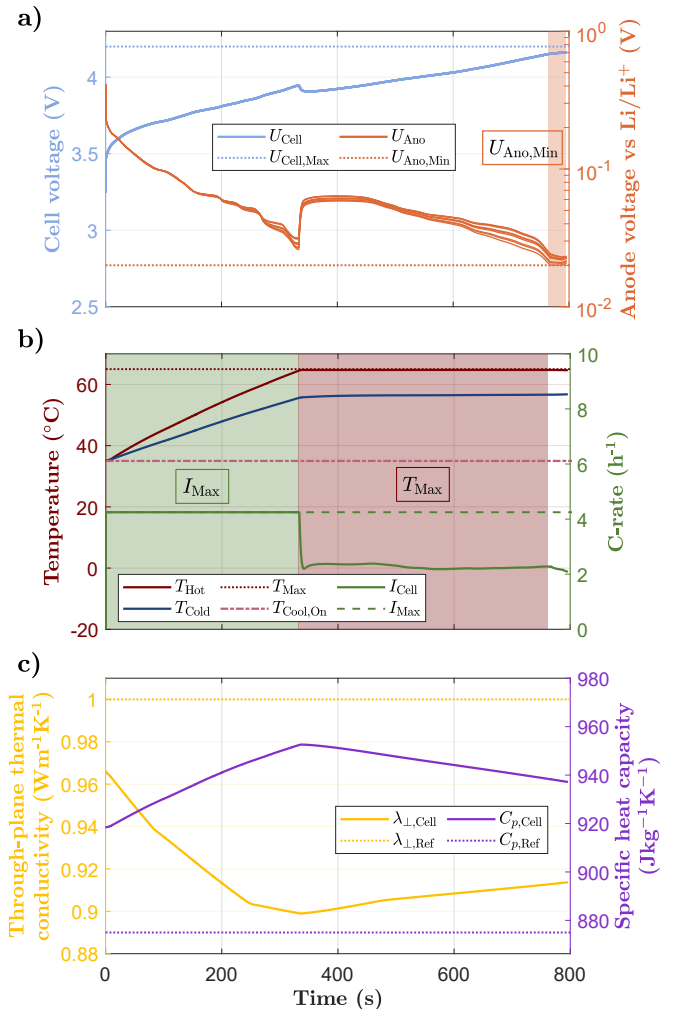


Fig. 3. Characteristic factors for an exemplary charging procedure from 10% to 80% SOC with bottom cooling at $T_{\text{Start}} = 35$ °C. (a) Cell voltage and anode voltage with corresponding limits. (b) Minimum and maximum cell temperature in combination with C-rate and corresponding limits. (c) Through-plane thermal conductivity and specific heat capacity compared to reference values.

limit between 330 s and 770 s. Finally, the minimum anode voltage of $U_{\text{Ano,Min}} = 20$ mV is reached at 770 s and kept constant in the final stage of the charging procedure. Preventing lithium-plating, a constantly decreasing charging current is applied for the remaining charging procedure until a SOC of 80% is reached after 795 s.

Due to the varying temperature and SOC throughout the charging procedure, the variable thermal parameter definition leads to a variation of through-plane thermal conductivity and specific heat capacity. As shown in Fig. 3(c), the through-plane thermal conductivity and specific heat capacity are plotted over time whereby the constant values of the reference case are depicted, too. On the one hand, the through-plane thermal conductivity is close to the constant reference value of $\lambda_{\perp} = 1$

$\text{Wm}^{-1}\text{K}^{-1}$ in the beginning. However, due to the rising cell temperature, the through-plane thermal conductivity steadily decreases for the first 330 s. Following, the cell temperature is nearly constant and the rising SOC leads to slight increase in through-plane thermal conductivity from $0.9 \text{ Wm}^{-1}\text{K}^{-1}$ to $0.915 \text{ Wm}^{-1}\text{K}^{-1}$. Although the rise in through-plane thermal conductivity at high SOC is beneficial for thermal homogeneity, its magnitude is small compared to the temperature-related decrease. On the other hand, the initial specific heat capacity is observably higher than the reference value of $C_p = 875 \text{ Jkg}^{-1}\text{K}^{-1}$. Moreover, the rising temperature leads to a further increase in specific heat capacity until a maximum of $C_p = 950 \text{ Jkg}^{-1}\text{K}^{-1}$ is reached after 330 s. Similar to the through-plane thermal conductivity, the following

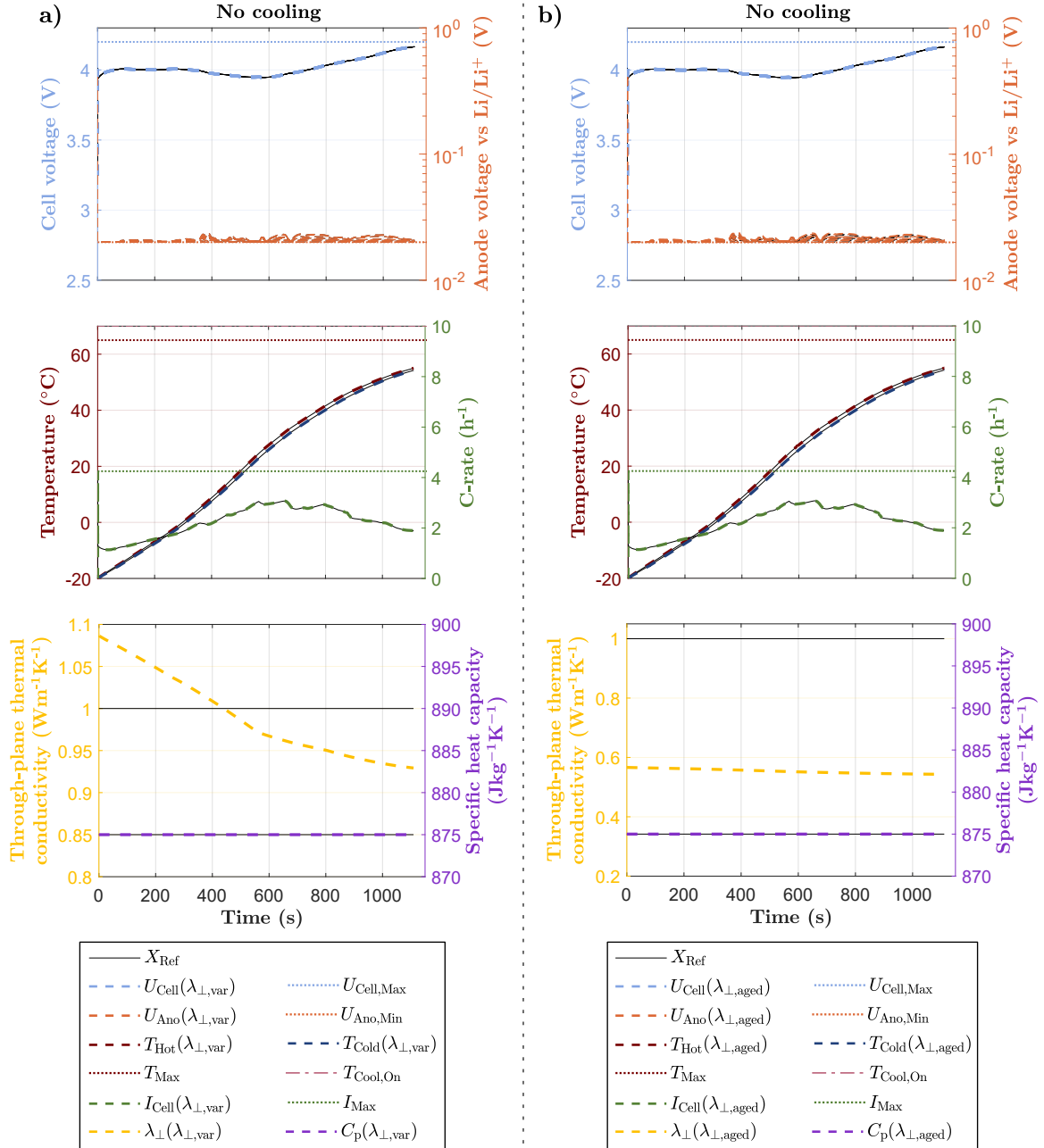


Fig. 4. Influence of variable through-plane thermal conductivity and aged through-plane thermal conductivity on fast-charging at cold starting temperatures. (a) Cell voltage, anode voltage, minimum and maximum cell temperature, C-rate, through-plane thermal conductivity and specific heat capacity with corresponding limits for constant and variable through-plane thermal conductivity for no cooling and $T_{\text{Start}} = 20$ °C. (b) Cell voltage, anode voltage, minimum and maximum cell temperature, C-rate, through-plane thermal conductivity and specific heat capacity with corresponding limits for constant and aged through-plane thermal conductivity for no cooling and $T_{\text{Start}} = 20$ °C.

decrease in specific heat capacity is linked to the rising SOC.

3.2. Influence of variable through-plane thermal conductivity on fast-charging

As previously demonstrated, the through-plane thermal conductivity of pristine lithium-ion cells depends on temperature and SOC. Therefore, corresponding dependencies are considered for different cooling scenarios and starting temperatures for the thermal simulation. Collected in Fig. 1(d) and (e), the linearly decreasing temperature dependency as well as the parabolic SOC dependency of the through-plane thermal conductivity are integrated into the thermal simulation and compared to the reference case of constant thermal parameters.

For low starting temperatures where no cooling is required, the implementation of a variable through-plane thermal conductivity does not measurably affect the fast-charging performance. Even though

temperature inhomogeneity is slightly reduced due to the increased through-plane thermal conductivity at low temperatures, the difference in through-plane thermal conductivity only amounts to less than 10% compared to the reference case. Furthermore, the non-existence of cooling results in a negligible effect of the variable through-plane thermal conductivity on fast-charging as observable in Fig. 4(a).

Similarly, a negligible effect of variable through-plane thermal conductivity can be found for bottom cooling. Visualized in Fig. 5(a), the charging procedure with bottom cooling and $T_{\text{Start}} = 0^\circ\text{C}$ is analyzed. It can be seen, that the variable through-plane thermal conductivity does not significantly influence the fast-charging performance for the underlying parameter set. Neither the voltage and anode voltage, nor the temperature and charging current observably differ between the reference and variable through-plane thermal conductivity case. As the variable through-plane thermal conductivity is close to the reference case, no significant effect on the thermal behavior is observed. In

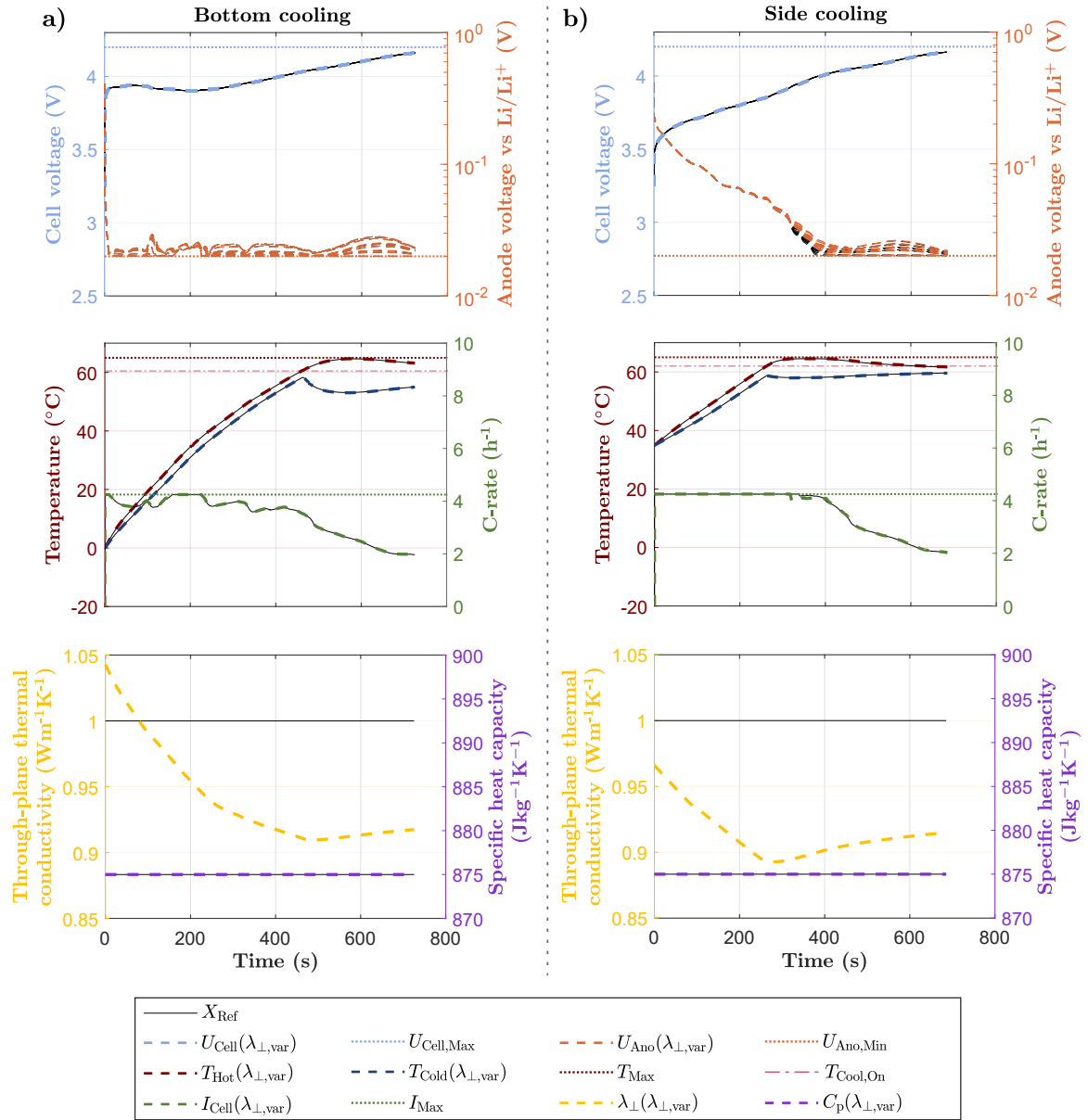


Fig. 5. Influence of variable through-plane thermal conductivity on fast-charging. (a) Cell voltage, anode voltage, minimum and maximum cell temperature, C-rate, through-plane thermal conductivity and specific heat capacity with corresponding limits for constant and variable through-plane thermal conductivity for bottom cooling and $T_{\text{Start}} = 0^\circ\text{C}$. (b) Cell voltage, anode voltage, minimum and maximum cell temperature, C-rate, through-plane thermal conductivity and specific heat capacity with corresponding limits for constant and variable through-plane thermal conductivity for side cooling and $T_{\text{Start}} = 35^\circ\text{C}$.

consequence, the charging time is equal and the variable parameter definition does not lead to a more precise description of the physical cell behavior. This is because the main cooling direction for bottom cooling is in axial direction and therefore mainly depends on the in-plane thermal conductivity. As this stays constant over the operating range, the moderate variation in through-plane thermal conductivity is insignificant.

Additionally, fast-charging with side cooling is also unaffected by a variable through-plane thermal conductivity definition for the analyzed sample cell. The charging procedure with a starting temperature of 35 °C and side cooling is shown in Fig. 5(b). Compared to the reference case with constant thermal parameters, a slight reduction in charging current is found at around 330 s. Since the high cell temperature leads to a moderately reduced through-plane thermal conductivity, the thermal inhomogeneity inside the cell only slightly increases. However, the described marginal differences between the reference and the variable through-plane thermal conductivity result in an increase in charging time of only two seconds.

In conclusion, the effect of a variable through-plane thermal conductivity definition is negligible for the analyzed sample cell. As the through-plane thermal conductivity only varies by about 10% during a charging procedure, no significant effects on the fast-charging performance result. While charging procedures with no cooling and bottom cooling inherently experience no significant dependency on through-plane thermal conductivity, even side cooling with the main heat dissipation in through-plane direction is nearly unaffected by the small variation in through-plane thermal conductivity. Therefore, the implementation of a constant through-plane thermal conductivity is reasonable for the analyzed sample cell as it does not lead to significant inaccuracies and precisely describes the thermal behavior of the lithium-ion cell. However, if cells with a significantly larger diameter are considered, the radial thermal inhomogeneity can increase. Furthermore, a stronger operating point dependence of the through-plane thermal conductivity could also lead to a measurably different thermal behavior for other lithium-ion cells. Whether other cells also do not require the consideration of a variable through-plane thermal conductivity thus depends on the cooling scenario, heat dissipation, cell diameter, through-plane thermal conductivity as well as the magnitude of the operating point dependence of the through-plane thermal conductivity. Furthermore, a significant effect of the variable through-plane thermal conductivity could be observed at extreme temperatures as observed during thermal runaway [38]. Although no significant effect for common automotive lithium-ion cells in common temperature regions is expected due to the low variation of the through-plane thermal conductivity and the limited cell diameter, parameter studies for various cell configurations have to be performed in future work to generalize found effects of the operating point-dependent through-plane thermal conductivity.

3.3. Influence of aged through-plane thermal conductivity on fast-charging

The through-plane thermal conductivity of lithium-ion cells also depends on the SOH. As described in Fig. 1(f), it can be reduced by up to 50% when heavily aged cells are considered. While the through-plane thermal conductivity of aged cells is strongly reduced, its temperature dependence significantly sinks and the SOC does not influence the through-plane conductivity of aged cells. Again, described dependencies are integrated into the thermal simulation and compared to the reference case of constant thermal parameters.

When the starting temperature is low and no cooling is required, the strongly reduced through-plane thermal conductivity only plays a negligible role. Even though temperature inhomogeneity is slightly increased, the non-existing cooling and low temperatures result in an unchanged fast-charging capability as described for the variable through-plane thermal conductivity, too. This can be observed in Fig. 4

(b).

Furthermore, the case of bottom cooling can lead to a small but observable influence of an aged through-plane thermal conductivity. Depicted in Fig. 6(a), the temperature delta between the inner-laying hot spot and the outer-laying cold spot is increased compared to the reference through-plane thermal conductivity. Thus, the temperature difference between the cooling plate and the cell is reduced when reaching critical temperature limits and the maximum cooling power decreases. Due to the reduced heat dissipation and increased self-heating of the cell, a temperature-related reduction in charging current is applied slightly earlier than for the reference case. Resulting in an increase in charging time of about 1%, the influence of the aged through-plane thermal conductivity for bottom cooling is limited.

However, the effect of an aged through-plane thermal conductivity on side cooling is evident, especially for high starting temperatures. Visualized in Fig. 6(b), the fast-charging time for $T_{\text{Start}} = 35\text{ °C}$ is significantly increased for the case of aged through-plane thermal conductivity. Being caused by the above described effect of increased temperature delta and reduced heat dissipation, the chosen cooling activation temperature of $T_{\text{Cool,On}} = 62\text{ °C}$ is insufficient for the case of the strongly reduced through-plane thermal conductivity. Since the charging current has to be suddenly reduced after 300 s for evading temperature limit exceedance, a lower cooling activation temperature would compensate the reduced heat transfer capability. Even though this effect can be corrected by a revised thermal management system (TMS) design with an adapted $T_{\text{Cool,On}}$, it emphasizes the importance of a customized TMS strategy with precise thermal parameters [39]. For application in battery electric vehicles this would lead to an operating point-dependent parametrization of the TMS which has to be updated according to the SOH and the magnitude of the through-plane thermal conductivity. Furthermore, the neglect of the aged through-plane thermal conductivity could also lead to a maximum temperature exceedance during operation which can then again increase the aging intensity. As this might result in a further reduced through-plane thermal conductivity, a negative feedback loop can result.

Summarizing, an aged through-plane thermal conductivity can have measurable effects on the fast-charging capability of the analyzed lithium-ion cell. Essentially reducing heat transfer and increasing the temperature inhomogeneity, the resulting reduced heat dissipation has adverse effects on charging time. In particular at high starting temperatures and for side cooling, a consideration of the aging-related decrease in through-plane thermal conductivity is beneficial and the TMS design has to account for the dependence of the through-plane thermal conductivity on SOH for the analyzed sample cell. Nevertheless, this work only considers a reduced aged through-plane thermal conductivity while not altering the electrical and electrochemical parameters. As for aged cells the increased inner resistance is of high importance as well, thorough analysis of aged cells with fully modified parameters including an aged through-plane thermal conductivity have to be conducted in future work. In addition, this work only considered the extreme case of halved through-plane thermal conductivity due to aging. Further analysis has to be conducted to analyze moderate through-plane thermal conductivity reduction at lower aging intensities. As described in the previous subsection, common automotive lithium-ion cells are supposed to be affected from the consideration of an aged through-plane thermal conductivity. While this depends on the cooling scenario, heat dissipation, cell diameter, through-plane thermal conductivity as well as the magnitude of the operating point dependence of the through-plane thermal conductivity, the effect of an aged through-plane thermal conductivity on fast-charging has to be analyzed for further sample cells.

3.4. Influence of variable specific heat capacity on fast-charging

Temperature significantly affects the specific heat capacity of lithium-ion cells. As described in Fig. 1(g), an increase of 0.25%/K is observed for the sample cell. Furthermore, the analyzed cell also has a

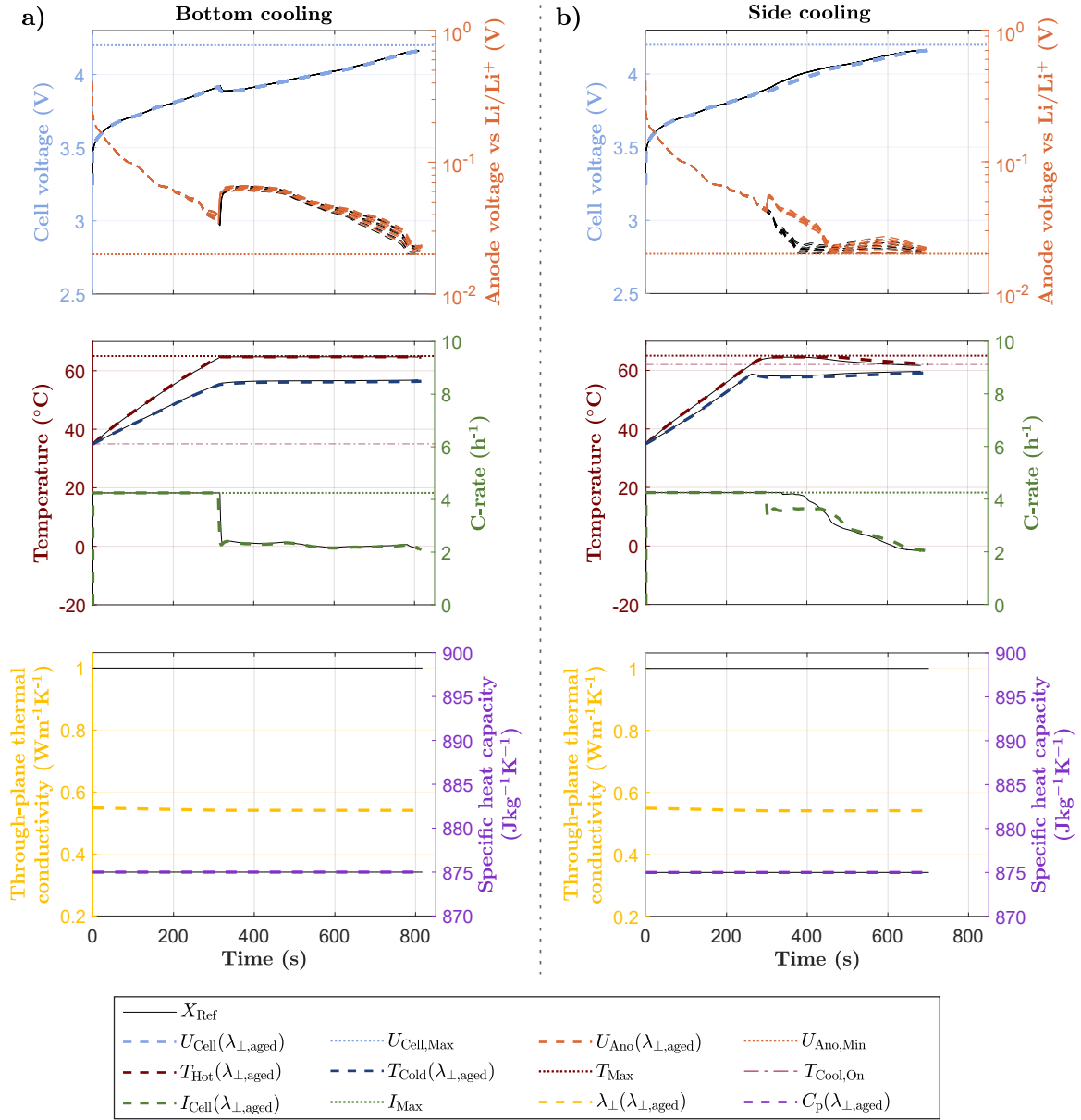


Fig. 6. Influence of aged through-plane thermal conductivity on fast-charging. (a) Cell voltage, anode voltage, minimum and maximum cell temperature, C-rate, through-plane thermal conductivity and specific heat capacity with corresponding limits for constant and aged through-plane thermal conductivity for bottom cooling and $T_{Start} = 35^\circ\text{C}$. (b) Cell voltage, anode voltage, minimum and maximum cell temperature, C-rate, through-plane thermal conductivity and specific heat capacity with corresponding limits for constant and aged through-plane thermal conductivity for side cooling and $T_{Start} = 35^\circ\text{C}$.

sinking specific heat capacity with rising SOC while aging does not have measurable effects on the specific heat capacity. Thus, the described parametric effects are implemented into the thermal modeling framework and resulting effects are assessed.

To begin with, fast-charging at low starting temperatures is significantly affected by a variable specific heat capacity. Visualized in Fig. 7 (a), the charging procedure with no cooling at $T_{Start} = 20^\circ\text{C}$ is illustrated for constant and variable specific heat capacity. Since the variable specific heat capacity is notably smaller as the reference specific heat capacity at low temperatures, the temperature of the variable specific heat capacity case rises faster and thus a higher charging current is applied. This again leads to an even faster temperature increase triggering a positive feedback loop to apply higher charging currents. Furthermore, the variable specific heat capacity exceeds the reference value of $C_p = 875 \text{ Jkg}^{-1}\text{K}^{-1}$ after 330 s resulting in a reduced temperature rise. Therefore, the final cell temperature of the variable specific

heat capacity case is very close to the cell temperature of the reference case. In consequence, the fast-charging time is reduced by about 3% due to the higher charging current for the variable specific heat capacity. Thus, the positive temperature dependence of the specific heat capacity has beneficial effects on fast-charging as it allows faster heating and shorter charging time.

Moreover, variable specific heat capacity also improves the fast-charging capability at high starting temperatures as shown in Fig. 7 (b). Because the variable specific heat capacity is constantly higher than the reference specific heat capacity for $T > 35^\circ\text{C}$, the cell temperature of the variable specific heat capacity case is measurably lower than the reference curve. Therefore, temperature related current reduction has to be performed later and the maximum charging current can be applied for a longer period. As a result, the charging time for the variable specific heat capacity case is reduced by 2% for the case of bottom cooling at $T_{Start} = 35^\circ\text{C}$.

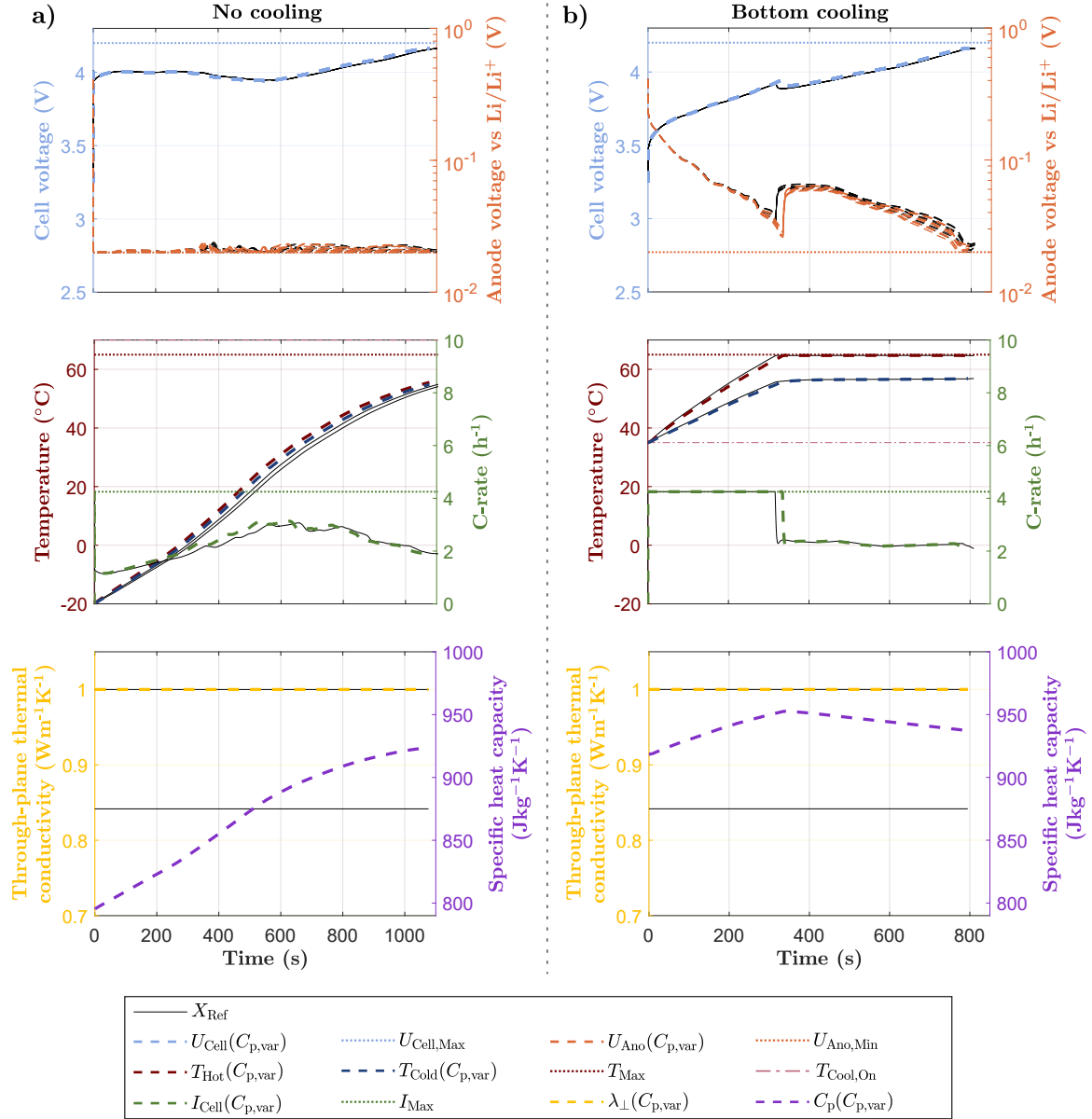


Fig. 7. Influence of variable specific heat capacity on fast-charging (a) Cell voltage, anode voltage, minimum and maximum cell temperature, C-rate, through-plane thermal conductivity and specific heat capacity with corresponding limits for constant and variable specific heat capacity for no cooling and $T_{Start} = 20$ °C. (b) Cell voltage, anode voltage, minimum and maximum cell temperature, C-rate, through-plane thermal conductivity and specific heat capacity with corresponding limits for constant and variable specific heat capacity for bottom cooling and $T_{Start} = 35$ °C.

Summarizing, the variable specific heat capacity definition has positive effects on fast-charging for the analyzed sample cell. While it improves the physical representation of the actual cell behavior, the positive correlation between temperature and specific heat capacity leads to an improved fast-charging capability at hot and cold temperatures. While the negative SOC dependency has adverse effects, its influence is not significant in any charging procedures due to the overcompensation of the positive temperature effect. Therefore, a variable specific heat capacity leads to a faster temperature rise and charging current increase at low temperatures while temperature-related charging current reduction at high temperatures is delayed. As this does not depend on the cooling scenario, heat dissipation and cell diameter, it is mainly affected by the specific heat capacity as well as the magnitude of the operating point dependence of the specific heat capacity. Since the specific heat capacity generally increases with rising temperature for common lithium-ion cells [18], the observed positive

effect on fast-charging time can be transferred to most automotive lithium-ion cells. Even though the charging time is only reduced by up to 3%, the effort for incorporating the dependence of specific heat capacity on the cell's operating point is limited. Because of this high benefit-cost ratio, the effect of the variable specific heat capacity on fast-charging is considered significant. Although precise magnitudes and detailed correlations for further cells have to be determined in future work, the integration of a variable specific heat capacity is important and should be carefully considered.

3.5. Influence of variable thermal parameters on fast-charging time

As extensively described in the previous subsections, the effects of variable thermal conductivity, aged thermal conductivity and variable specific heat capacity significantly on the fast-charging time of lithium-ion cells strongly differ. Nevertheless, they are not mutually exclusive

and thus their interdependence requires thorough analysis, too. In consequence, the following subsection gives a comprehensive overview of various combinations of cooling scenario, starting temperature and thermal parameter set. Linear interpolation between the individual points is reasonable due to the approximately linear correlation between

starting temperature and charging time in the analyzed temperature region as found by Pegel et al. [24].

Fig. 8(a) illustrates the dependency of fast-charging time on thermal parameters for low starting temperatures $T_{\text{Start}} \leq 10^\circ\text{C}$ with no cooling. It can be seen that the charging time for colder starting temperatures is

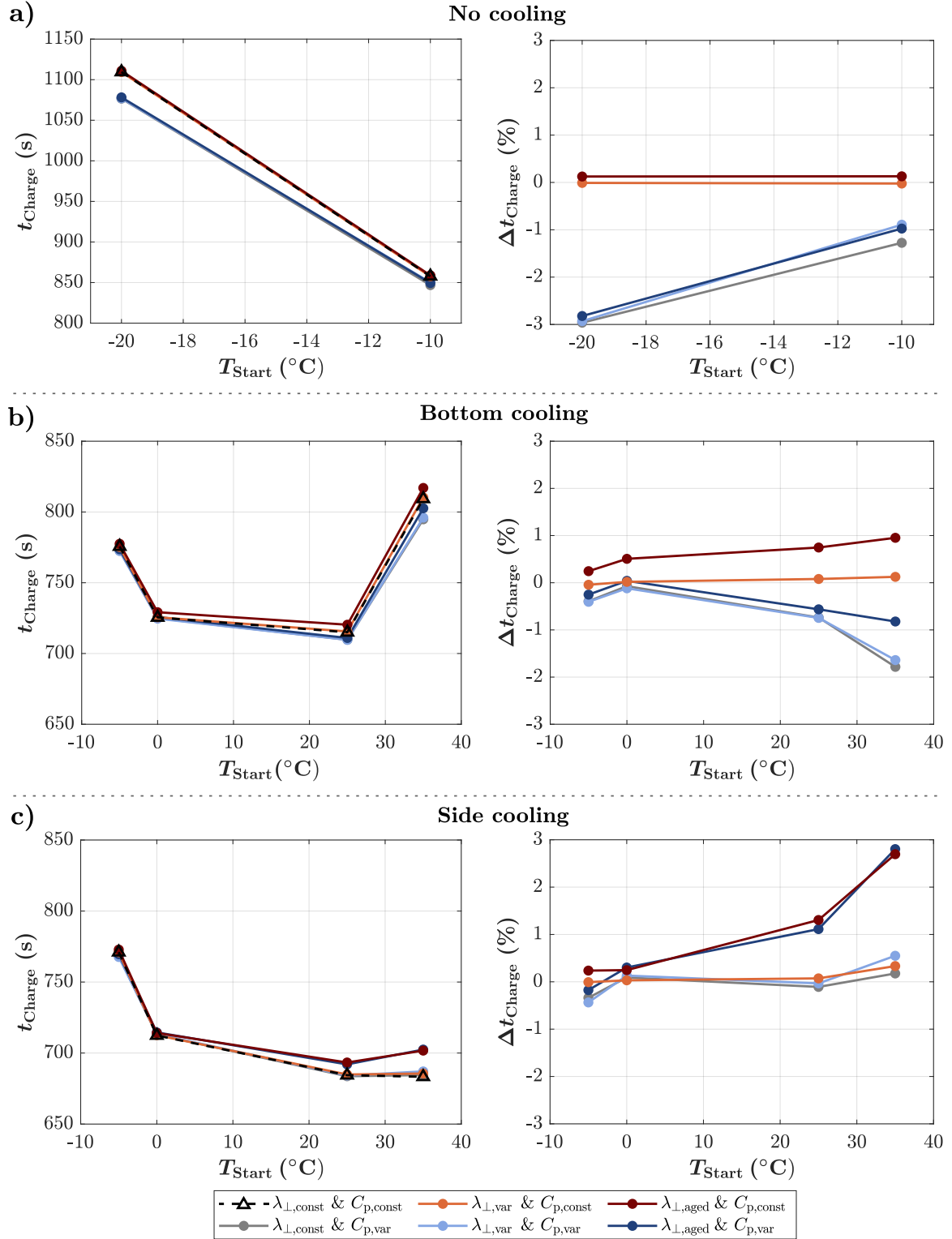


Fig. 8. Comprehensive effect of variable thermal parameters on fast-charging time for all cooling scenarios and starting temperatures. (a) Fast-charging time for no cooling and relative change in fast-charging time resulting from different thermal parameter sets. (b) Fast-charging time for bottom cooling and relative change in fast-charging time resulting from different thermal parameter sets. (c) Fast-charging time for side cooling and relative change in fast-charging time resulting from different thermal parameter sets.

higher than for room temperature. Being 1110 s and 858 s for $T_{\text{Start}} = 20^\circ\text{C}$ and $T_{\text{Start}} = 10^\circ\text{C}$ for the reference case, respectively, the charging times for the variable specific heat capacity are significantly lower. Reducing the charging time by 3% and 1%, the variable specific heat capacity allows for a higher charging current due to the faster temperature increase as described in the previous subsection. However, neither variable through-plane thermal conductivity nor aged through-plane thermal conductivity play a significant role at low starting temperatures. As shown in Fig. 8(a), fast-charging times for thermal parameters involving variable or aged through-plane thermal conductivity are identical to the reference case of constant thermal parameters. This is explained by the non-existing cooling and the resulting minor importance of radial heat transfer with no cooling.

Next, the charging times for bottom cooling at starting temperatures between $T_{\text{Start}} = 5^\circ\text{C}$ and $T_{\text{Start}} = 35^\circ\text{C}$ are illustrated in Fig. 8(b) with various thermal parameter definitions. The U-shaped curve of the charging time is the result of the temperature-dependent charging current. While the anode potential limits the maximum charging current due to the cold temperature at $T_{\text{Start}} = 5^\circ\text{C}$, the insufficient cooling power at $T_{\text{Start}} = 35^\circ\text{C}$ leads to a temperature-related charging current reduction and a higher charging time. The variable through-plane thermal conductivity has no effect on the charging time for bottom cooling. Because the cooling is applied in the in-plane direction, a moderate variation in through-plane thermal conductivity does not affect the fast-charging time. Furthermore, the aged through-plane thermal conductivity has a linearly increasing effect on the fast-charging time with increasing temperature. Even though being limited, the strongly reduced through-plane thermal conductivity increases the difference between the hot spot and cold spot which reduces the cooling power and heat dissipation. Therefore, the initially chosen TMS design involving a nonideal $T_{\text{Cool,On}}$ is not ideal anymore and a small increase in charging time of 1% is observed. Moreover, the variable specific heat capacity does not have significant effects on the charging time between $T_{\text{Start}} = 5^\circ\text{C}$ and $T_{\text{Start}} = 25^\circ\text{C}$ due to the moderate temperatures. Since the variable specific heat capacity is similar to the reference case in the underlying average temperature regions, the effect on charging time is limited. However, at $T_{\text{Start}} = 35^\circ\text{C}$ a measurable effect of the variable specific heat capacity is visible. Because the heat capacity at high temperatures is higher for the variable specific heat capacity as for the reference case, more heat can be stored inside the cell. Thus, the temperature related charging current reduction can be performed later and the maximum charging current is applied longer, resulting in a reduction in charging time of up to 2%. In addition, it is demonstrated that the combination of variable thermal parameters behaves as the superposition of the individual effects. When the positive effect of a variable specific heat capacity is combined with a negative impact of the aged through-plane thermal conductivity, the change in charging time is roughly the sum of both effects. For example at $T_{\text{Start}} = 35^\circ\text{C}$, the aged-through-plane thermal conductivity leads to an increase in charging time of 1%. While the variable specific heat capacity leads to a reduction in charging time of 2%, the consideration of both thermal parameter definitions results in a decrease in charging time of 1%. As this can be observed for all data points, the implementation of multiple variable thermal parameters is possible and leads to physically reasonable results.

Finally, Fig. 8(c) depicts the charging times for various starting temperatures with side cooling. The above described U-shaped curve of charging time is not observed for side cooling due to the high cooling power. Thus, $T_{\text{Start}} = 35^\circ\text{C}$ does not lead to a higher charging time compared to room temperature because no temperature-related current reduction has to be applied. Similar to bottom cooling, the variable through-plane thermal conductivity does not significantly affect the fast-charging time for side cooling, too. As the cooling power is high, the slight reduction in through-plane thermal conductivity is negligible. In contrast, the aged through-plane thermal conductivity shows a distinct impact on the fast-charging time. Especially for higher starting

temperatures, the increased thermal inhomogeneity leads to a reduced cooling power and thus the chosen TMS design is imperfect. Again emphasizing the importance of a customized TMS design adapted to tailored thermal parameters, the charging time for a starting temperature of $T_{\text{Start}} = 35^\circ\text{C}$ can increase by up to 3% if the aged through-plane thermal conductivity is considered. Furthermore, the variable specific heat capacity has no beneficial effects on the fast-charging time with side cooling for $T_{\text{Start}} \geq 5^\circ\text{C}$. As the faster temperature increase for charging with low starting temperatures is only limited for $T_{\text{Start}} = 5^\circ\text{C}$ and $T_{\text{Start}} = 0^\circ\text{C}$, only a minor effect of the variable specific heat capacity can be observed. Similarly, only a negligible effect is found for high starting temperatures. As the cooling power for side cooling is very high, the increased thermal budget does not decrease the charging time because no temperature-related current reduction is applied in the reference case, too. Again, it is evident that the combination of variable thermal parameters shows the superimposed impact of the individual effects. Even though only aged through-plane thermal conductivity shows significant effects on the fast-charging time, the superposition of aged through-plane thermal conductivity and variable specific heat capacity at $T_{\text{Start}} = 5^\circ\text{C}$ and $T_{\text{Start}} = 35^\circ\text{C}$ shows a combined effect of the individual influences.

4. Conclusion

This work thoroughly characterized the influence of operating point-dependent thermal parameters on fast-charging of lithium-ion cells first-ever. Integrating the experimentally determined dependence of thermal conductivity and specific heat capacity on temperature, SOC and SOH, a well tested and validated thermal-electrical-electrochemical modeling framework by Pegel et al. [24] was employed.

It was found that the variable through-plane thermal conductivity definition does not significantly affect the thermal behavior of the analyzed lithium-ion cell compared to the reference case of constant thermal parameters. As the through-plane thermal conductivity only varies by about ten percent over the full operating range of the cell, the fast-charging time was only negligibly influenced by a variable parameter definition. In contrast, the integration of the aged through-plane thermal conductivity showed a measurable influence on fast-charging of the sample cell. As the extreme case of a strongly aged cell with a halved through-plane thermal conductivity was chosen, thermal transport in the through-plane direction was impaired which can increase the charging time by up to 3%. Furthermore, a variable definition of the specific heat capacity notably influences fast-charging of lithium-ion cells. Because the operating point-dependent specific heat capacity significantly differs in extreme temperature regions, a lower specific heat capacity at cold temperatures leads to a faster increase of temperature allowing for a higher charging current and reduction in charging time of 3%. In addition, high starting temperatures resulted in a higher specific heat capacity compared to the reference case increasing the thermal budget of the cell while reducing charging time.

Because found effects of the operating point-dependent through-plane thermal conductivity strongly depend on the cooling scenario, heat dissipation, cell diameter, through-plane thermal conductivity value as well as the magnitude of the operating point dependence of the through-plane thermal conductivity, presented influences of the through-plane thermal conductivity on the fast-charging of the sample cell cannot be generalized. However, the narrow range of through-plane thermal conductivity values and limited cell diameters for common automotive lithium-ion cells should not result in extreme effects of an operating point-dependent through-plane thermal conductivity. In contrast, found beneficial effects of the variable specific heat capacity on fast-charging are expected to be transferable to most common lithium-ion cells. Due to the positive correlation between specific heat capacity and temperature, most lithium-ion cells benefit from a variable specific heat capacity definition. Thus, it is highly recommended to consider the operating point-dependent specific heat capacity for

thermal simulations due to the high benefit-cost ratio and wide-ranging applicability. The physical behavior of lithium-ion cells is thereby emulated more precisely improving fast-charging capability. However, found results need to be analyzed for further sample cells and experimental validation has to be performed.

This work assessed the influence of operating point-dependent thermal parameters on the fast-charging of lithium-ion cells for the first time. Significant correlations were identified and consequences for thermal simulation and thermal management design were presented. Even though many dependencies were determined, further thermal simulation frameworks and various sample cells need to be analyzed in future work. Furthermore, experimental validation of found simulation results has to be performed and the influence of measurement uncertainty for the thermal parameters has to be analyzed.

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CRediT authorship contribution statement

Luca Tendra: Writing – original draft, Visualization, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Hendrik Pegel:** Writing – review & editing, Validation, Supervision, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Carlos Gonzalez:** Writing – review & editing, Visualization, Validation, Software, Methodology, Investigation, Data curation. **Dominik Wycisk:** Writing – review & editing, Methodology, Investigation, Conceptualization. **Alexander Fill:** Writing – review & editing, Methodology, Formal analysis. **Kai Peter Birke:** Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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